

NexusFin

Problem Statement 3: Responsible Credit and Informed Choice

Helping borrowers and lenders understand what credit is realistically affordable before committing.

Technical Product + Modular API Architecture • Solo Applicant: Edwin Mwiti • October 2026

NexusFin is a responsible credit decision-support platform that transforms raw financial data and proposed loan terms into understandable affordability insights, multi-scenario stress tests, and standardized Key Facts comparisons — without relying on opaque black-box credit scores.

Credit access is not the same as credit affordability

Expanding digital credit without affordability assessment creates severe debt-trap risks across ASEAN.

- Volatile Informal Incomes: 48% of Filipino adults say raising emergency funds is very difficult; gig and MSME workers face fluctuating monthly cash flows (BSP CFIS 2026).
- Hidden Debt Concentration: Existing debts and recurring family obligations reduce available disposable cash flow, leading to compounding debt rollover.
- Misleading Flat-Rate Marketing: Borrowers often evaluate headline instalments without understanding the total cost of credit or effective APR.
- Shock Vulnerability: A repayment that appears manageable in a peak month becomes unsustainable when an unexpected income drop or medical emergency occurs.

Core Opportunity: Make affordability, total costs, and financial resilience trade-offs transparent before committing to credit.

NexusFin: Explainable credit decision support for borrowers and lenders

A platform designed as transparent decision support — not an opaque automated approval/rejection machine.

1. Understand	Capture declared or consented income, essential living costs, existing debt, and liquid reserves.
2. Assess Affordability	Calculate exact post-loan cash buffer, debt-service burden (DTI), and total financing costs.
3. Stress Test	Simulate 6 realistic shock scenarios (10%, 25%, 40% income drops, inflation surges, and dual shocks).
4. Compare & Explain	Generate side-by-side Key Facts Statements and plain-language trade-offs rather than black-box scores.

From loan question to informed, resilient choice

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|-----------|---|
| 01 | Financial Health Profile
Income + volatility + living expenses + existing debt + liquid reserves |
| 02 | Credit Offer Input
Principal + term + APR + upfront fees + repayment structure (amortizing vs flat) |
| 03 | Affordability Engine
Deterministic post-loan monthly buffer & debt-service burden calculation |
| 04 | Shock Stress Testing
Simulate 6 life shocks to verify if cash flow remains positive |
| 05 | Standardized Key Facts
Side-by-side comparison of options & plain-language trade-off guidance |

Useful financial signals without invasive surveillance

NexusFin leverages consented alternative transaction data while enforcing strict privacy by design.

- Privacy by Design: ZERO contact-list scraping, zero social-media surveillance, zero invasive phone permission grabs.
- Financial Relevance Only: Ingests consented transaction cash flows (bank, mobile wallet, or merchant turnover) and utility consistency.
- Pattern Recognition: Classifies irregular gig/client earnings, estimates income volatility (CV), and cross-checks declared liabilities.
- Deterministic Core Math: Safety-critical calculations are 100% deterministic code (zero AI black-box in arithmetic). NLP is strictly used to translate complex actuarial results into plain, localized language.
- Data Governance: Granular opt-in/opt-out toggles with declared legal basis and governance audit trail.

Making financial stress visible before it becomes a crisis

Borrower Profile	Carlos — Manila Food & Delivery Gig Rider (PHP)
Monthly Income	PHP 38,000 (average with 25% estimated variability)
Essential Living Costs	PHP 21,000 (rent, groceries, utilities, fuel)
Existing Debt Payments	PHP 4,500 (motorcycle installment)
Proposed Credit Offer	PHP 35,000 over 10 months at 24% APR (PHP 4,046.43/mo payment with fees)
Base Case Buffer	PHP 8,454/month positive buffer (Debt Burden: 22.5% — Manageable baseline)
25% Income Shock Test	Income drops to PHP 28,500 -> Buffer becomes -PHP 1,046/mo (DEFICIT)

The NexusFin Difference: Rather than discovering insolvency mid-loan, NexusFin concludes: *Manageable today, vulnerable under income shock*. Carlos sees the 25% shock deficit in advance and receives guidance to negotiate a longer term or lower principal to preserve resilience.

Modular, containerized, and Open Finance ready

Designed for rapid pilot deployment with financial institutions and regulatory sandboxes.

Frontend SPA	Responsive dashboard built with semantic HTML5, modern CSS design system, and vanilla JS (zero heavy framework overhead).
FastAPI Backend	High-performance Python asynchronous modular REST API architecture with Pydantic v2 strict typing and OpenAPI specifications.
Calculation Engines	Deterministic loan amortization, fee accounting, 6-scenario stress testing, and evidence-based Multidimensional Resilience Overview.
Alternative Data	Automated CSV transaction ingestion, classification keyword rules, and volatility coefficient analysis.
Governance & Audit	Granular consent registry with governance audit log stream aligned with BSP Circular No. 1133.
Deployment	Fully containerized via Docker and docker-compose; ready for cloud or on-premise deployment.

Designing for measurable financial-health outcomes

Moving beyond access metrics to verifiable household resilience.

Over-Indebtedness Prevention	Measure reduction in borrowers taking credit with >40% debt burden or negative shock buffers.
Informed Choice Comprehension	Measure user understanding of total financing charges and effective APR across competing products.
Resilient Credit Expansion	Enable microfinance partners to underwrite thin-file gig workers through verified cash flows.
Pilot Cohort Metrics	Track assessment completion, loan right-sizing frequency, and 90-day delinquency rate vs control cohorts.

Viable B2B SaaS business model with regional replicability

- B2B SaaS / API Tier: Financial institutions, rural banks, and cooperatives pay per assessment API call to improve responsible underwriting and ESG compliance.
- Employer & Gig Platform Channel: Delivery platforms (Grab, Foodpanda) and employers integrate NexusFin as a worker financial wellness perk.
- Free Neutral Consumer Tool: Core borrower workspace remains 100% free and independent from predatory referral commissions.
- Modular ASEAN Localization: Pre-calibrated for all ASEAN currencies (PHP, IDR, SGD, MYR, THB, VND, KES, USD) with adaptable central bank regulatory thresholds.
- Pilot-First Strategy: Designed for immediate entry into the BSP Regulatory Sandbox and partner microfinance clinics.

From working prototype to ASEAN-wide implementation

Phase 1 (Months 1–2)	Validation in BSP Regulatory Sandbox with Philippine microfinance cooperative or digital bank.
Phase 2 (Months 3–4)	Controlled pilot with 1,000 gig and MSME borrowers; measure loan right-sizing and user comprehension.
Phase 3 (Months 5–6)	Longitudinal evaluation of 90-day loan performance; calibrate country-specific volatility thresholds.
Phase 4 (Months 7+)	Regional expansion into Indonesia (OJK framework), Vietnam (SBV), and Thailand (BOT).

NexusFin — Making credit decisions easier to understand before they become financial burdens.

ASEAN Financial Health Challenge • Problem Statement 3: Responsible Credit and Informed Choice • Team NexusFin